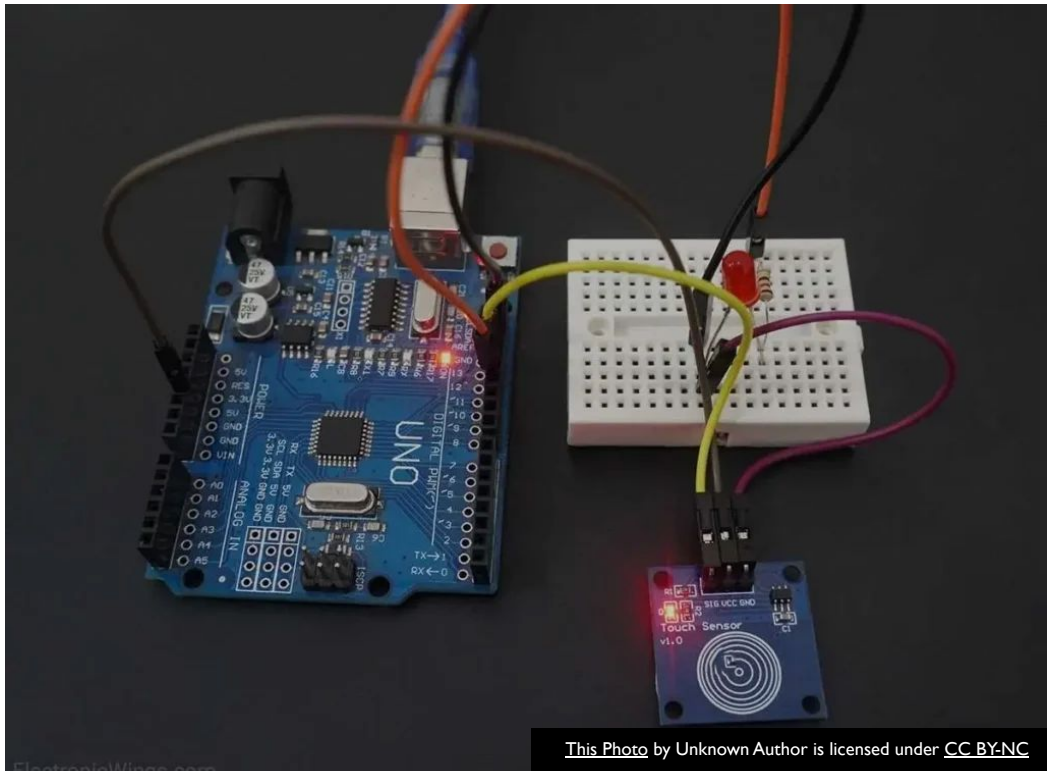
SENSORS: THE SENSES OF A MACHINE

HOW ROBOTS "FEEL" THE WORLD AROUND THEM



WHAT IS A SENSOR?



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- A sensor is a device that **detects a change in its environment** and converts it into a signal (usually electrical) that a system can understand.
- Think of it like this:
  - Your eyes = light sensor
  - Your skin = touch/temperature sensor
  - Your ears = sound sensor
- **A sensor is basically an artificial sense organ for a machine.**

WHY DO ROBOTS NEED SENSORS?

- Without sensors, a robot is just blindly executing code — it has no idea what's actually happening around it.
- If a robot arm doesn't have a touch sensor, what could go wrong when it tries to pick up an egg?
- Sensors give robots **feedback**, and feedback enables **decision-making**.

CLASSIFICATION OF SENSORS

1. Based on Power Requirement

Active sensors (need external power, e.g., ultrasonic sensor)

Passive sensors (generate signal on their own, e.g., thermocouple)

2. Based on Output Type

Analog sensors (continuous signal, e.g., LDR light sensor)

Digital sensors (discrete on/off signal, e.g., IR touch sensor)

CLASSIFICATION OF SENSORS

3. Based on Function

Physical sensors (light, temperature, pressure)

Chemical sensors (gas, pH)

Biosensors (used in medical robotics)

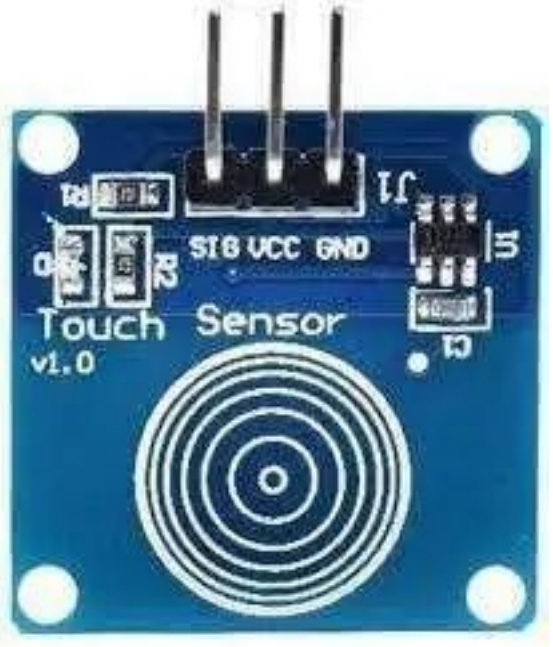
4. Based on Contact

Contact sensors (touch, pressure)

Non-contact sensors (light, IR, ultrasonic)

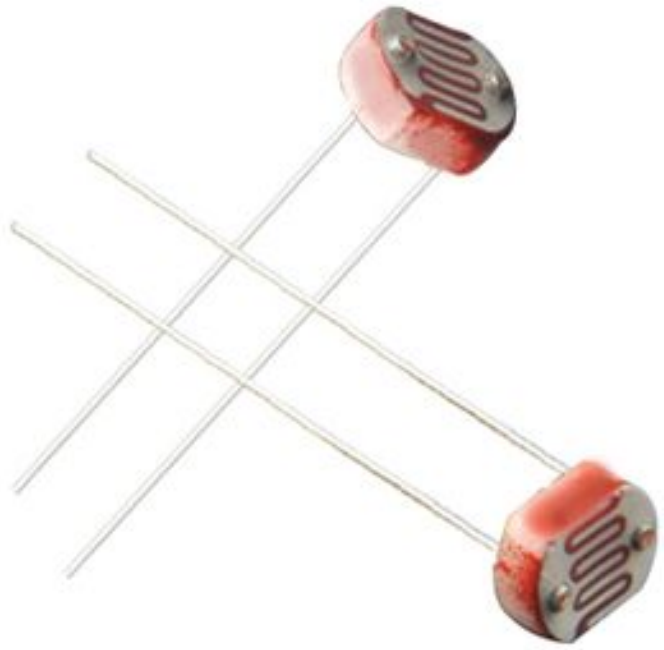
EXAMPLE 1 — TOUCH SENSOR

- **What it does:** Detects physical contact/pressure.
- **Simple analogy:** Like a doorbell — press it, circuit completes, signal sent.
- **Robotics application:**
  - Robotic grippers (know when they've grasped an object)
  - Bumper sensors on mobile robots (obstacle detection)
  - Humanoid robots (basic sense of "touch" for safe interaction)



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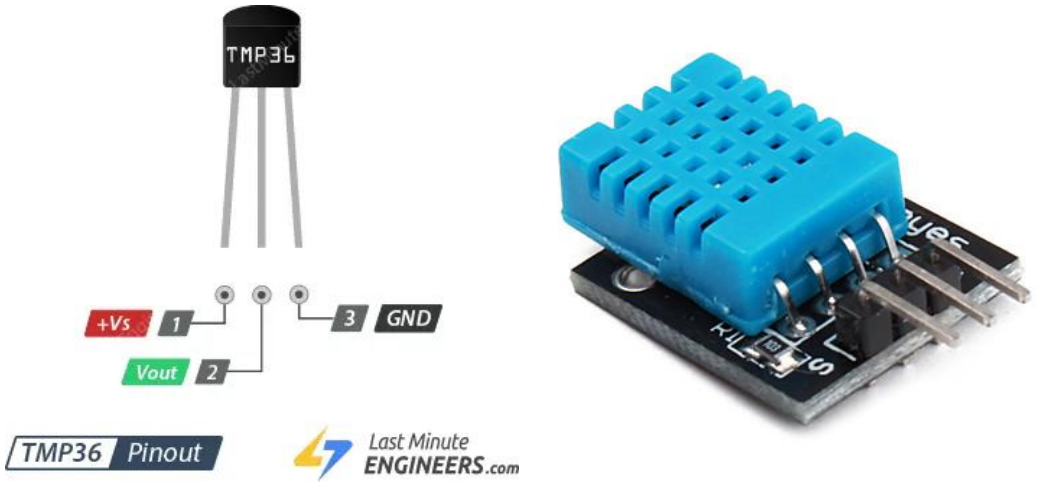
EXAMPLE 2 — LIGHT SENSOR



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- **What it does:** Detects intensity of light (e.g., LDR – Light Dependent Resistor).
- **Simple analogy:** Like your eyes squinting in bright sunlight.
- **Robotics application:**
  - Line-following robots (detect black line vs white surface)
  - Solar-tracking robots
  - Automatic lighting systems in smart environments

EXAMPLE 3 — TEMPERATURE SENSOR

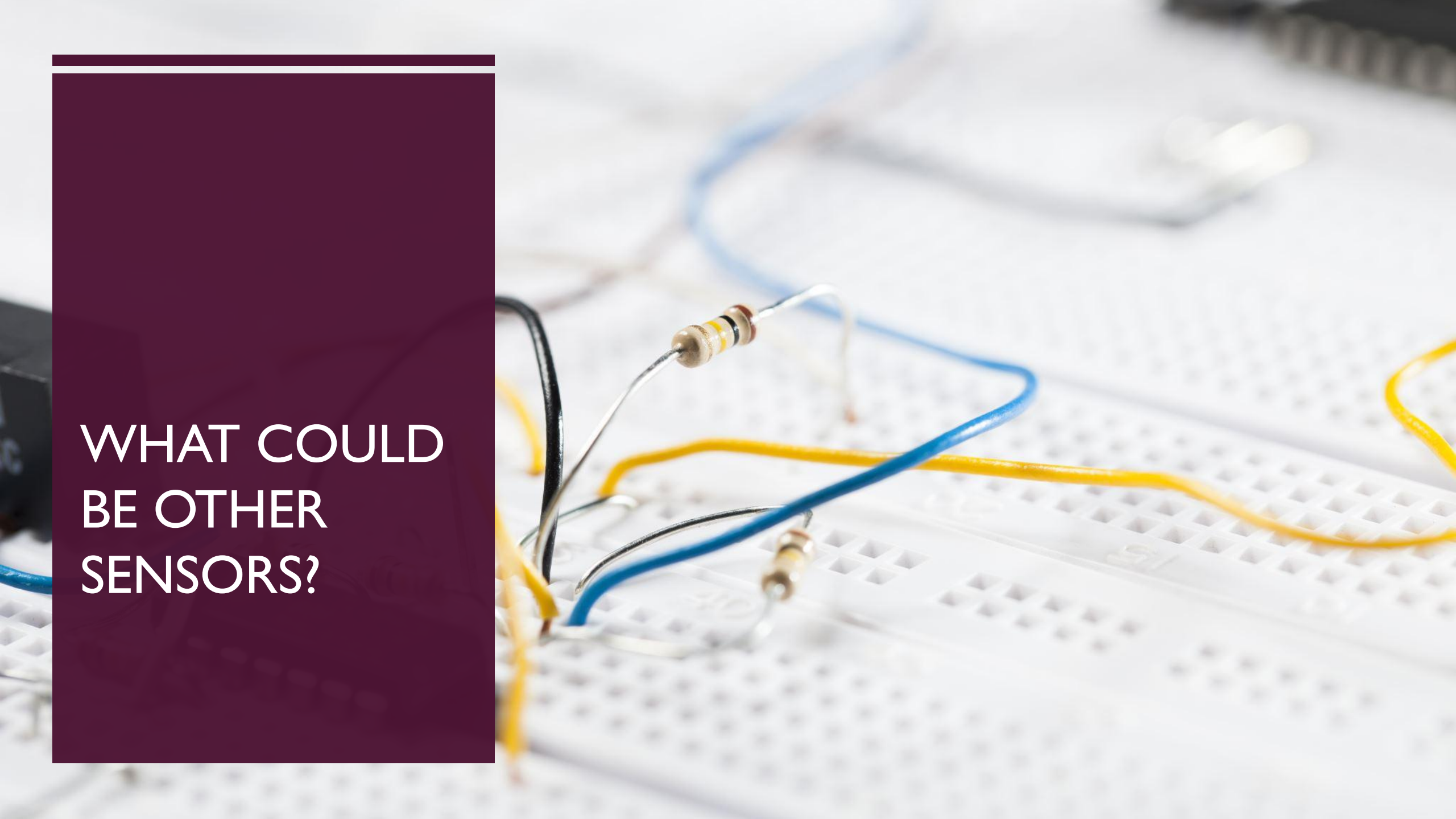


- **What it does:** Detects heat/cold levels (e.g., thermistor, LM35, DHT11).
- **Simple analogy:** Like touching a cup to check if your tea is too hot.
- **Robotics application:**
  - Fire-fighting robots
  - Industrial robots monitoring motor overheating
  - Weather-monitoring drones/robots

QUICK COMPARISON TABLE

Sensor	Detects	Output Type	Robotics Use
Touch	Contact/Pressure	Digital/Analog	Grippers, bumpers
Light	Light Intensity	Analog	Line following, navigation
Temperature	Heat level	Analog	Safety, fire detection

WHAT COULD BE OTHER SENSORS?



COMMON SENSORS — QUICK REFERENCE

Sensor	Definition
Proximity Sensor (IR/Capacitive)	Detects the presence of a nearby object without physical contact.
Ultrasonic Sensor	Measures distance using reflected sound waves.
Infrared (IR) Sensor	Detects objects or measures distance/heat using infrared radiation.
Pressure Sensor	Measures force applied per unit area (e.g., air or liquid pressure).
Gas Sensor (MQ series)	Detects the presence/concentration of specific gases in the air.
Humidity Sensor	Measures the amount of water vapor in the air.
Accelerometer	Measures acceleration and orientation/tilt of an object.
Gyroscope	Measures angular velocity to detect rotation or orientation change.
IMU (Inertial Measurement Unit)	Combines accelerometer + gyroscope (often + magnetometer) for motion/orientation sensing.

COMMON SENSORS — QUICK REFERENCE

Sensor	Definition
Sound Sensor (Microphone)	Detects sound intensity or specific audio signals.
Color Sensor	Detects and identifies the color of an object's surface.
Magnetic Sensor (Hall Effect)	Detects presence/strength of a magnetic field.
Flame/Fire Sensor	Detects presence of flame or fire via infrared light.
PIR (Passive Infrared) Sensor	Detects motion by sensing changes in infrared radiation from a moving body.
LiDAR Sensor	Measures distance/mapping using reflected laser light.
GPS Sensor	Determines geographic location using satellite signals.
Encoder	Measures the rotational position or speed of a shaft/motor.
pH Sensor	Measures the acidity or alkalinity of a solution.

HOW WILL YOU DECIDE IF A SENSOR IS GOOD OR POOR?

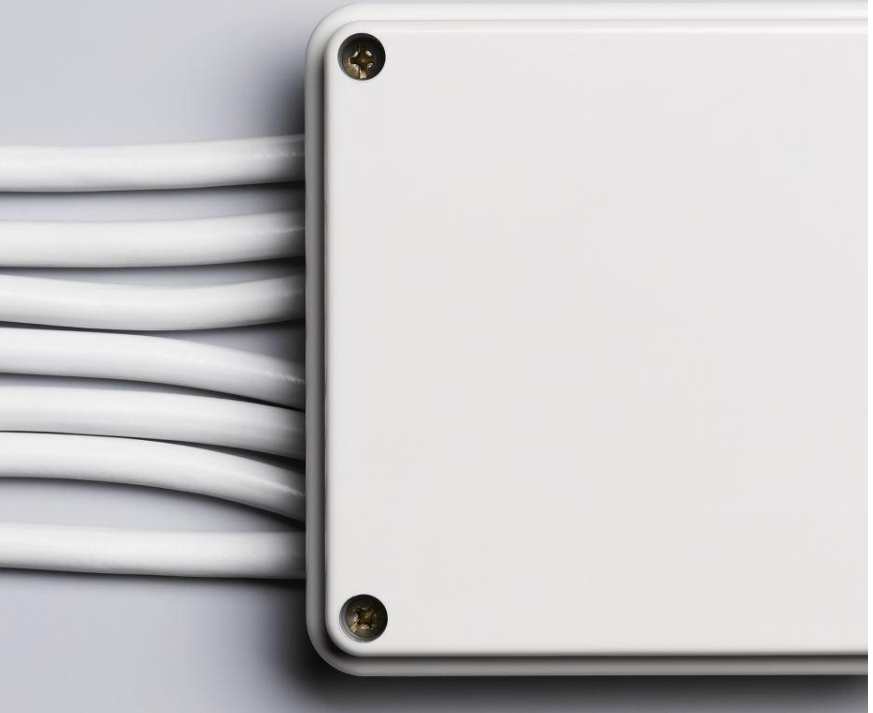
CHOOSING A SENSOR

Criteria	Question to Ask
Accuracy	How close is the reading to the true value?
Precision	Does it give consistent readings every time?
Response Time	How fast does it detect a change?
Range	What's the minimum/maximum it can measure?
Resolution	What's the smallest change it can detect?
Cost & Power	Does it fit the project budget and power supply?
Environment	Will it survive heat, dust, water, vibration?

CAN WE USE A SIMPLE LM35 TEMPERATURE SENSOR IN A CAR ENGINE?

INDUSTRIAL SENSORS VS. HOBBYIST SENSORS

Aspect	Industrial Sensors	Hobbyist Sensors
Build Quality	Rugged, sealed, shock/vibration resistant	Lightweight, often exposed PCB
Accuracy/Calibration	High precision, factory-calibrated	Moderate, often needs manual calibration
Operating Range	Wide (extreme temp, pressure, humidity)	Limited to normal lab/room conditions
Durability/Lifespan	Built for continuous 24/7 operation, years of use	Short-medium lifespan, breadboard-level use
Communication	Industrial protocols (Modbus, RS-485, CAN, 4-20mA)	Simple digital/analog, I2C, UART
Cost	High ($50 – $1000+)	Low ($1 – $20)
Certification	IP-rated, ISO/CE certified for safety-critical use	No formal certification
Example	Siemens pressure transmitter, Sick optical sensor	LM35, HC-SR04, LDR m



PROS & CONS

- **Industrial Sensors**
  - Pros: Extremely reliable, long-term stability, safe for hazardous/critical environments
  - Cons: Expensive, overkill for learning/prototyping, harder to interface without industrial controllers (PLC)
- **Hobbyist Sensors**
  - Pros: Cheap, easy to prototype with (Arduino/ESP32 plug-and-play), great for learning
  - Cons: Lower accuracy, not built for harsh environments, shorter lifespan, less repeatable results

DIP VS. SMD SENSOR PACKAGES

Feature	DIP (Dual In-line Package)	SMD (Surface Mount Device)
Mounting	Inserted through holes, soldered on board	Soldered directly onto board surface
Size	Larger, bulkier	Very small, compact
Assembly	Easy hand-soldering, breadboard-friendly	Needs precision tools/reflow soldering
Repairability	Easy to replace/desolder	Difficult to rework by hand
Cost at Scale	Higher (more material, manual labor)	Lower (mass production, automated placement)
Common Use	Prototyping, teaching labs, hobby projects	Commercial products, mobile devices, compact PCBs
Example	DIP-packaged LM35, breadboard IR sensor modules	SMD LM75 temp sensor, MEMS accelerometers in phones



PROS AND CONS

- **DIP**
  - Pros: Beginner-friendly, easy to prototype/debug, replaceable without special equipment
  - Cons: Bulky, not practical for compact/mass-produced electronics
- **SMD**
  - Pros: Space-saving, ideal for compact robotics (drones, wearables), cheaper in mass production
  - Cons: Hard to solder/replace by hand, needs specialized tools, harder for beginners

QUICK TAKEAWAY

Industrial vs Hobbyist = a *reliability & environment* trade-off

DIP vs SMD = a *size & manufacturability* trade-off

Both decisions come down to: **What's the application, and what constraints (cost, space, durability) matter most?**